SAROJ PRASAD CHHATOI

+33-749254998

spchhatoi@laas.fr
chhatoisaroj@gmail.com

Education

- ***LAAS-CNRS, Toulouse, France***
  Automatic Control — Feb 2023 - March 2026
  Doctoral Candidate
  Supervisors: Aneel Tanwani, Didier Henrion
- ***University of California, Los Angeles***
  Noncommutative Optimal Transport — Mar. 2025 - Jun. 2025
  Visiting Graduate Researcher
- ***Institute Of Mathematical Sciences, Chennai***
  Physics — Aug. 2017 - Jul. 2020
  Master of Science, Junior Research Fellow
- ***Indian Institute Of Technology Jodhpur***
  Systems Engineering with Specialization in Electrical Engineering — Jun. 2013 - May 2017
  Bachelor of Technology

Journal Articles

Evolution of Measures in Nonsmooth Dynamical Systems
Co-authors: Aneel Tanwani, Didier Henrion
Automatica, (Vol. 179, art. 112402, 2025)

Optimizing Quasi-Dissipative Evolution Equations with the Moment-SOS Hierarchy
Co-authors: Swann Marx, Nicolas Seguin, Didier Henrion
Discrete and Continuous Dynamical Systems – Series A (DCDS-A)(Vol. 45, no. 11, 2025)

Approximating the Order-2 Quantum Wasserstein Distance Using the Moment–SOS Hierarchy
Co-author: Victor Magron
Optimization Letters, (Vol. 20, Issue 1, 2026)

Optimal Control For Articulated Soft Robots
Co-authors: Michele Pierallini, Franco Angelini, Carlos Mastalli, Manolo Garabini
IEEE Transactions on Robotics (Vol. 39, No. 5, Oct. 2023)

Inverse Dynamics based Model Predictive Control via Nullspace Resolution
Co-authors: Carlos Mastalli, Thomas Corbères, Steve Tonneau, Sethu Vijayakumar
IEEE Transactions on Robotics (Vol. 39, No. 4, Aug. 2023)

Unpublished Articles and Preprints

Optimal Control of State Constrained Systems via Measure Relaxations and Polynomial Optimization
(Under review, with Aneel Tanwani and Didier Henrion)

A Gaussian surrogate of partially observed stochastic processes using Wasserstein metric
(Under review, with Ibrahim Ramadan and Aneel Tanwani)

Quantum Gradient Structures from Large Deviations and Quantum Thermodynamics
White paper, IPAM Noncommutative Optimal Transport long program (co-author, with participants).

Shannon- and von Neumann-entropy regularizations of linear and semidefinite programs

(with Jean B. Lasserre)

Conference Articles

Optimal Control of Nonsmooth Dynamical Systems using Measure Relaxations
- IEEE Conference on Decision and Control 2024, Milan (`https://hal.science/hal-04701838`)
Co-authors: Aneel Tanwani, Didier Henrion

Articulated Soft Robots: A Feasibility Driven Control Approach
- I-RIM Conference 2024, Rome (DOI: 10.5281/zenodo.14731033)
Co-authors: Davide De Benedittis, Michele Pierallini, Franco Angelini, Carlos Mastalli, and Manolo Garabini

Work Experiences

Centro E. Piaggio, University of Pisa, Italy
- Research Fellow Jul. 2021 - Jan. 2023

- Optimal planning and control of legged robots.

Flytbase Inc., Pune, India
- Robotics Engineer Apr. 2021 - Jun. 2021

- Part of the precision landing team.

HiTech Robotic Systems Ltd., Gurgaon, India
- Senior Research Engineer Dec. 2020 - Apr. 2021

- Responsible for improving the existing control and motion planning architecture.

Robotics Research Center, International Institute of Information Technology, Hyderabad, India
- Research Intern under Dr. K. Madhava Krishna Mar. 2020 - Nov. 2020

- Worked on lab's funded project with Rockwell Collins Aerospace on non-holonomic multiagent motion planning frameworks.

Workshops and Graduate Courses

- Oberwolfach workshop on "Polynomial Optimization for Nonlinear Dynamics" (Aug. 2024)
- Stochastic Calculus (Graduate Course, IMT Toulouse, Instructor: Serge Cohen)(Sept. 2024 - Nov. 2024)
- Elliptic PDEs and Evolution equations (Graduate Course, IMT Toulouse, Instructor: Mihai Maris)(Sept. 2024 - Nov. 2024)
- Introduction to Optimal Transport (Graduate Course, IMT Toulouse, Instructor: Jerome Betrand)(Sept. 2024 - Nov. 2024)
- Convergence of Probability measures (Graduate Course, IMT Toulouse, instructor: Clement Pellegrini)(Sept. 2023 - Nov. 2023)
- Stochastic processes (Graduate Course, IMT Toulouse, Instructor: Francois Chapon)(Jan. 2024 - Apr. 2024)
- Summer school on Applied Harmonic Analysis and Machine learning, Genoa (Aug. 2024)
- Summer school on Nonsmooth dynamical systems, Paris (Apr. 2023)

Awards

- **Graduate Fellow**, Institute for Pure and Applied Mathematics, University of Los Angeles, California (Mar. 2025 - Jun. 2025)
- **Leibniz graduate fellow**, Oberwolfach Workshop on "Polynomial Optimization for Nonlinear Dynamics: Theory, Algorithms, and Applications", Mathematisches Forschungsinstitut Oberwolfach (Aug. 2024)
- **Student Travel and Workshop Support** for IEEE Conference on Decisions and Control 2024
- **Best poster presentation** at IEEE-RAS TC Model-Based Optimization for Robotics 2023

"Inverse Dynamics MPC via Nullspace Resolution"

- **Finalists for best paper award** at IEEE-ICRA TC Model Based Optimization for Robotics 2024 "Inverse Dynamics MPC via Nullspace Resolution"

Skills

Languages: C, C++, Julia, Python, MATLAB, Fortran, Mathematica
Tools/Platforms: ROS, Gazebo, Git, LaTeX
Libraries/Frameworks: NumPy, SciPy, scikit-learn, TensorFlow, PyTorch, GloptiPoly
Certifications: Coursera (2020): Intro to TensorFlow for AI/ML/DL (PEJJL4QPWX9K), CNNs in TensorFlow (Q5AXUBEQ835C), Sample-based Learning Methods (GPLL62V6R36N)

Referees

- **Tryphon T. Georgiou** `tryphon@uci.edu`
- **Marius Junge** `mjunge@illinois.edu`
- **Didier Henrion** (Supervisor) `henrion@laas.fr`
- **Aneel Tanwani** (Supervisor) `tanwani@laas.fr`